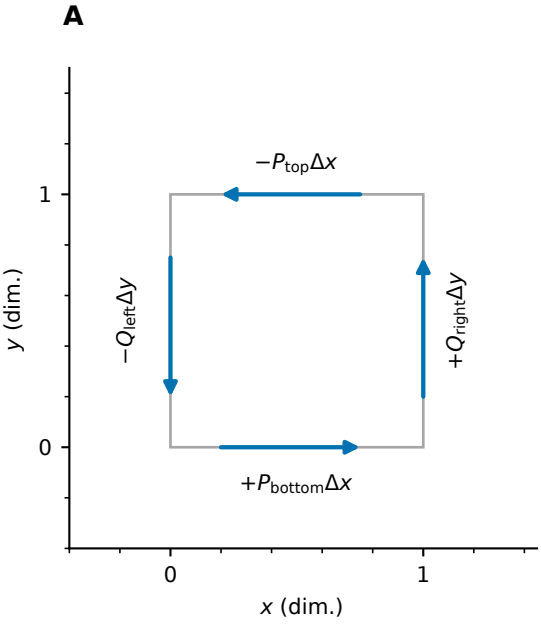
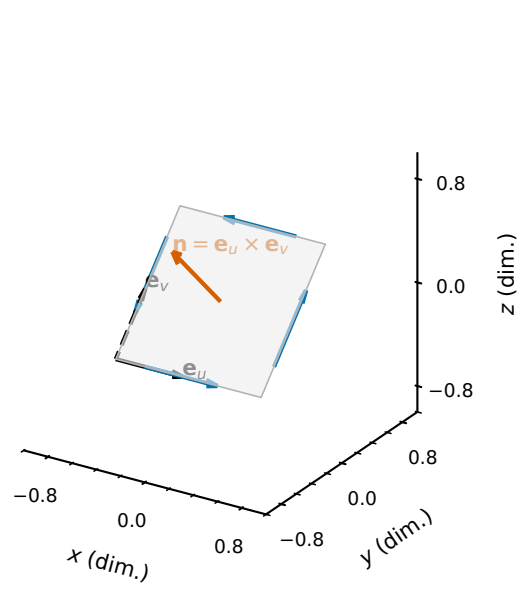


CCW circulation: $(Q_x - P_y)\Delta x \Delta y$



Oriented area: $\mathbf{n}\,dS$ (right-hand rule)